Department	Компьютердик инженерия				
Program	Магистратура ☒	Докторантура ☐			
Program Name	Компьютердик инженерия, магистратура				
Semester	III☒	IV☐	V ☐	VI☐	VII☐
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Dissertation Title	Neural network for diagnosing heart diseases using 12-channel electrocardiography (ECG) based on signal augmentation with the reference node method.
Short Content	
Keywords	SVG, ECG, CNN, Machine Learning, cardiovascular diseases, electrocardiogram, 12-lead signals, ECG interpretation

1. General Information about the Dissertation Topic

According to the World Health Organization (WHO), cardiovascular diseases (CVDs) are the leading cause of death globally, with approximately 18 million people dying each year. Cardiovascular diseases

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include coronary artery disease, cerebrovascular disease, rheumatic heart disease, and other conditions of the heart and blood vessels. One-fifth of all deaths from heart disease and stroke occur before the age of 70. However, accurate and early diagnosis can significantly reduce treatment costs and sudden cardiac death.

An electrocardiogram (ECG) is one of the simplest and fastest tests, which helps identify some heart conditions by measuring the heart's electrical activity.
Any disturbance in heart rhythm or conduction, or damage to the heart muscle, can change the heart's electrical activity. An abnormal ECG can also be a sign of a past heart attack or a high risk of having a heart attack in the near future.

The three main types of ECG are:

- Resting ECG: You lie down for this type of ECG. No movement is allowed during the test as electrical impulses from other muscles may interfere. This type of ECG usually takes 5 to 10 minutes.
- Ambulatory ECG: For an ambulatory ECG, you wear a portable recording device for at least 24 hours. You can move around normally while the device is attached.
- Stress Test/Exercise Stress Test: This involves having an ECG while riding a stationary bicycle or walking on a treadmill. Completing this type of ECG takes 15-30 minutes. The stress test may also involve medications to understand their effect on the heart.

Some heart problems that can cause changes in an ECG test include:

- Atrial fibrillation/flutter
- Heart attack (myocardial infarction)
- Heart failure
- Congenital heart defects
- Abnormal heart rhythms (arrhythmias)
- Poor blood supply to the heart
- Multifocal atrial tachycardia
- Paroxysmal supraventricular tachycardia
- Sick sinus syndrome
- Wolff-Parkinson-White syndrome

The accuracy of an ECG depends on the condition being tested. Heart disease may not always be visible on an ECG. Some heart conditions never cause any specific ECG changes.

Deep Neural Networks (DNNs) are widely used for classification and prediction tasks in various domains. Recently, it has been observed that DNNs have been sharply developed and have a significant impact on classification accuracy for a wide spectrum of medical tasks.

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(Electrocardiogram, 2024; Services, n.d.)
(Electrocardiogram Information | Mount Sinai - New York, n.d.)
(Ayal et al., 2023)
(Ebrahimi et al., 2020)

2. Problem Addressed by the Dissertation Topic

Automated 12-lead ECG devices can perform a general screening and provide a second opinion to physicians. However, correct interpretation of ECG recordings is complex and clinically challenging; incorrect interpretation can lead to poor outcomes or even patient death. Despite its widespread use, several studies have highlighted deficiencies in ECG interpretation skills among healthcare workers. Furthermore, interpretation can vary between different observers, and even for a single observer, while the task is monotonous and time-consuming even for highly experienced cardiologists. As literature shows, up to 33% of ECG interpretations have some error compared to expert reference, and up to 11% have led to incorrect management. There is no universal foundation in the pedagogy of ECG interpretation; time allocation, trainer development, and teaching format vary widely in the literature.

(Classification of 27 Heart Abnormalities Using 12-Lead ECG Signals with Combined Deep Learning Techniques | Rawt | Bulletin of Electrical Engineering and Informatics, n.d.)
(Breen et al., 2019)

3. Objective of the Dissertation Work

To develop and implement a neural network for diagnosing heart diseases using 12-channel electrocardiography (ECG) data based on signal augmentation using the reference node method. The development of a neural network for diagnosing heart diseases based on the analysis of 12-channel electrocardiography (ECG) is considered. This includes data collection, data preprocessing, model development and training, and evaluating its effectiveness.

4. Significance/Benefits of the Dissertation Work

Advances in machine learning present a potential opportunity to enhance the comprehensive interpretation of ECG. Many studies demonstrate the ability of DNN-ECG algorithms not only to detect various arrhythmias from a single-lead ECG but also to provide human-like interpretation of 12-lead ECGs. Furthermore, signal interpretation is a tedious and complex task, so even for experts trained over many years, there can be subjective variability and human error in the analysis process. Therefore, it is of great importance to research and develop computerized methods. It should assist healthcare professionals in making decisions that reduce the likelihood of errors and reduce the time spent diagnosing complex ECGs.

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The project of creating a neural network for diagnosing heart diseases through 12-lead ECG and signal augmentation methods is significant and beneficial for several reasons:

1. Early Diagnosis: Using an automated system to analyze ECG data enables the detection of heart disease signs at an early stage when treatment is most effective.

2. Reduced Diagnostic Time: An automated system can significantly reduce the time required for ECG analysis compared to manual interpretation by a specialist. This can be particularly useful in cases where a fast and accurate diagnosis is required.

3. Improved Accuracy: Using neural networks and signal augmentation techniques, a model can be created that can automatically learn the signs of diseases and make predictions with high accuracy, helping to prevent errors common in manual interpretation.

4. Improving Healthcare Accessibility: The automated system can be easily integrated into clinical practice and used in various healthcare institutions, improving the accessibility of cardiovascular disease diagnosis and treatment for patients.

5. Improved Treatment Outcomes: Faster and more accurate diagnosis of heart disease can lead to earlier treatment and improved treatment outcomes for patients, ultimately improving their quality of life.

Thus, the project can significantly improve the efficiency and accessibility of heart disease diagnosis, which is important for healthcare and prolonging patients' lives.

(Rafie et al., 2021)
(Liu et al., 2021)

5. Scope and Limitations of the Dissertation Work

DNNs can achieve high accuracy in detecting various cardiac anomalies, often surpassing traditional methods, and in some cases, even human experts. They can also help in early detection of conditions like arrhythmia, myocardial infarction, and other heart diseases, helping to improve the patient's condition. However, DNNs require large volumes of labeled data for training, which is difficult to obtain due to privacy concerns and the need for expert annotation. Deep learning models may not generalize well to unseen data or different patient populations, leading to potential inaccuracies.

The following methods are planned to be used in the research:

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- Signal augmentation technique
- Reference node method (SVM).

Using the signal augmentation technique allows improving the quality of training data and increasing the model's accuracy. This increases the diversity of training examples and improves the model's generalization ability. Using the Support Vector Machine (SVM) method allows obtaining accurate results for classification and regression. The experiment will be conducted based on data provided by the "Bicard" cardiology clinic.
The research includes the following main steps:

1. Data preparation
2. Model development
3. Training and testing
4. Results evaluation
5. Documentation preparation and presentation of results

(Ribeiro et al., 2020)

6. Hypotheses and Assumptions

1. DNNs can distinguish complex patterns from raw ECG data that are difficult for humans to detect.
2. DNNs can improve the accuracy of ECG interpretation with its large dataset.

The project has significant practical importance because automating the diagnosis of heart diseases based on ECG greatly improves the medical care process. The originality of the project lies in its unique approach to diagnosing heart diseases. The proposed method is expected to capture any changes in the ECG signal, thereby increasing diagnostic accuracy and improving patient treatment outcomes.

1. Use of 12-lead ECG: The project proposes using data from 12-lead ECG for diagnosing heart diseases. This allows capturing a wide spectrum of the heart's electrical activity, thus increasing the informativeness for classification.
2. The signal augmentation technique using the reference node method allows generating new variants

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of ECG data.

3. Use of neural networks for diagnosing heart diseases: neural networks can analyze data and make predictions with high accuracy, which can be useful for diagnosing various types of heart diseases.

4. Integration of machine learning algorithms into clinical practice: If the project is successful, it could lead to the development of an automated heart disease diagnostic tool integrated into clinical practice for faster and more accurate diagnosis and treatment of patients.

Thus, the novelty of the project lies in the use of modern techniques for augmentation, deep learning, and analysis of ECG signals for automated diagnosis of heart diseases. This leads to improved quality of healthcare and reduced burden on medical staff.

7. Methodological Basis of the Dissertation Work

7.1. Conceptual Foundations/Research Model

1. Data Preparation:
    1. Obtaining a 12-lead ECG dataset, including recordings from patients with various heart diseases and normal ECGs.
    2. Data preprocessing, such as amplitude normalization, noise removal, time series alignment, artifact removal, etc.
2. Data Augmentation:
   Developing a signal augmentation technique using the reference node method to create new versions of ECG data, accounting for various physiological variations and artifacts.
    1. Use signal augmentation on the prepared dataset to create an augmented dataset.
3. Neural Network Architecture Development:
    1. Definition of the neural network architecture for classifying ECG data.
    2. Selection of layer types (e.g., convolutional layers, pooling layers, fully connected layers) and

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their parameters.

3. Define activation functions, loss functions, and metrics for evaluating model performance.

4. Model Training:

   1. Splitting data into training, validation, and test sets.

   2. Training the model on the training data using the selected neural network architecture and signal augmentation technique.

   3. Evaluate the model's performance on the validation dataset and adjust model parameters if necessary.

5. Model Evaluation:

   1. Evaluate the performance of the trained model on the test dataset.

   2. Compare results with other methods of heart disease classification or diagnosis.

   3. Analyze model errors and explore its interpretability.

7.2. Population (Target Population) and Sample (Sample Population)

PhysioNet/Cinc 2020 challenge dataset and a dataset of hospital patients.

7.3. Materials and Their Collection

This work will use the PhysioNet/Cinc 2020 challenge dataset. This is a publicly available, multi-class ECG signal dataset containing 43,101 ECG recordings. Furthermore, the dataset is collected from various sources, making it ideal for ensuring the generalization ability of the proposed method.

(4 Large Scale 12-Lead Electrocardiogram Database for Arrhythmia Study v1.0.0, n.d.)

7.4. Data Analysis (Analysis and Summary of Materials)

The source dataset consists of ECGs from 45,152 patients. The number of A/D conversions per bit is 4.88,

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and the A/D converter had a 32-bit resolution. The amplitude unit was microvolts. The upper limit was 32,767, and the lower limit was -32,768. The Institutional Review Boards of Shaoxing People's Hospital and Ningbo First Hospital approved this study and allowed the waiver of informed consent and the public sharing of data after de-identification.

8. Work Plan/Schedule

Steps	Time								
	2025/2026-academic year Fall Semester					2025/2026-academic year Spring Semester			
	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Review of Available Research									
Research and Implementation									
Research and Implementation									
Research and Implementation									
Testing									
Validation									
Report Preparation									

9. Sources

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10. Appendix
11. Other Additional Items (Information about permission from the Ethics Committee or other official permission for writing the dissertation)
This dissertation project proposal was written in accordance with the rules specified in the KTMU, KII, Master's/PhD program instructions for writing and publishing a dissertation, as well as the instructions for writing a dissertation project proposal.

Signature Master's Student Nazarkulov Elaman 02.09.2024	Signature Scientific Supervisor Assoc. Prof. Rayımbek SULTANOV 02.09.2024

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